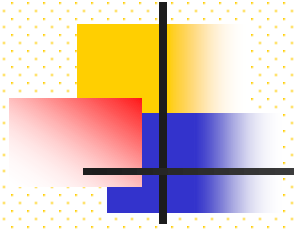


Polyuria and Polydipsia

Endocrinology

2025

Prof Karen Lam (Medicine)



Diabetes mellitus

A chronic disorder characterised by raised blood glucose levels, secondary to a complete or relative lack of insulin. Affects 10.5% adults worldwide in 2021 (Diabetes Res Clin Pract 2022).

SIGNS AND SYMPTOMS

THE THREE "P's"



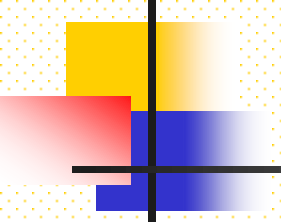
POLYURIA



POLYDIPSIA



POLYPHAGIA



Presentation

1. Asymptomatic – 50% - incidental glycosuria or hyperglycaemia
2. Classical symptoms – polyuria; polydipsia; weight loss despite increased appetite (marked weight loss suggests more severe diabetes or presence of TB)
3. Presenting with complications
4. Unmasked by steroid therapy, infection, pregnancy, stroke
5. Others – e.g. pruritus vulvae

Diagnosis of Diabetes: WHO/IDF/ADA 2025



Based on Venous Plasma Glucose

Fasting glucose ≥ 7 mmol/L (126 mg/dl; fasting ≥ 8 hours)

OR

2-hour glucose ≥ 11.1 mmol/l (200 mg/dl) during
75 gm Oral Glucose Tolerance Test (OGTT)

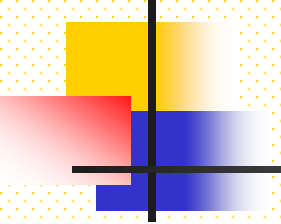
OR

Random plasma glucose ≥ 11.1 mmol/l in presence of
classical symptoms of diabetes or hyperglycaemic crisis

Based on HbA1c: HbA1c $\geq 6.5\%$

Must be based on standardised HbA1c assays with stringent quality assurance

Diagnosis of DM requires two abnormal test results from the same sample (eg. fasting glucose & HbA1c) or in two separate test samples, unless clearly symptomatic or hyperglycaemic crisis



Prediabetes (ADA) Intermediate hyperglycaemia (WHO)

By Glucose Criteria

IFG (Impaired Fasting Glucose)

Fasting glucose 6.1 to 6.9 mmol/L (WHO)

5.6 to 6.9 mmol/L (ADA)

OR

IGT (Impaired Glucose Tolerance)

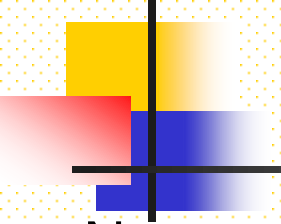
2-hour glucose ≥ 7.8 to <11.1 mmol/L during 75 gm OGTT

*IGT/IFG carry an increased risk of developing
diabetes and cardiovascular diseases*

By HbA1c criteria – ADA: 5.7-6.4%

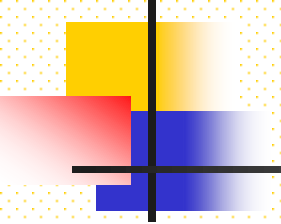
Canada/Europe: 6-6.4%

WHO: no HbA1c criteria for Prediabetes



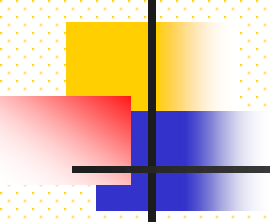
Glycosylated haemoglobin (HbA1c)

- Non-enzymatic glycosylation of N-terminal valine of Hb β -chain
 - Clinically used as an objective index of overall blood glucose control in the preceding 2-3 months
 - Good correlation with average blood glucose
- Exceptions:
1. Variable effect on HbA1c: Genetic/chemical alterations in Hb
 2. Falsely high HbA1c: $\uparrow$ glycation (eg. acidosis in chronic renal failure); $\downarrow$ erythropoiesis (eg iron deficiency);
 3. Falsely low HbA1c: $\downarrow$ red cell life span (eg. blood loss; haemolysis; hypersplenism; chronic renal failure); blood transfusion



Serum fructosamine (glycosylated serum proteins)

- Objective index of overall blood glucose control in the preceding 1-3 weeks.
- Correct for serum albumin level (if < 3.0 mmol/L).



Metabolic Syndrome

("The Insulin Resistance Syndrome")

Cluster of metabolic disorders linked by insulin resistance
& associated with accelerated atherosclerosis

Glucose intolerance/type 2 DM

Hypertension

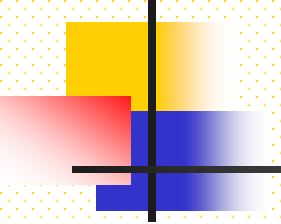
Hypertriglyceridemia

↓ HDL cholesterol

Central obesity

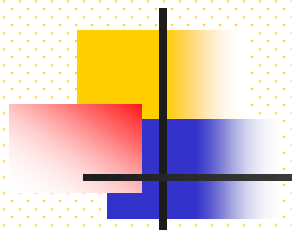
Diagnosis of Metabolic Syndrome: presence of ≥ 3 of the above

(Alberti KG et al, Circulation 2009)



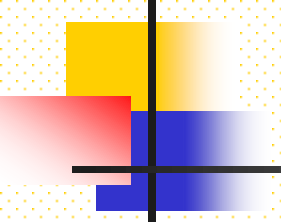
Classification

- Type 1: β -cell destruction, mostly autoimmune in etiology: life-long insulin dependency.
- Type 2: reduced insulin secretion & sensitivity: diet $\pm$ oral drugs $\pm$ insulin
- Other specific types:
 - LADA: Latent Autoimmune Diabetes in Adults- slow but progressive autoimmune β -cell destruction; years of marginal insulin secretion; type 2 DM with difficult control; other autoimmune diseases; non-obese; +ve islet autoantibodies; ?Type 1 (under debate)
 - Monogenic diabetes eg maturity-onset diabetes of the young (MODY; onset $\leq$ 25 years of age; autosomal dominant; >14 known genes: MODY1/2/3 most common)
 - Secondary diabetes: pancreatic diseases, endocrine diseases, drugs (eg. glucocorticoids, immunosuppressants) may remit if underlying cause removed
- Gestational diabetes



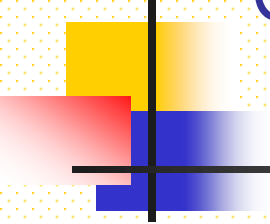
GDM (gestational diabetes)

1. 14% of pregnancies globally (IDF Atlas 2022)
2. Transient diabetes / impaired glucose tolerance during pregnancy
3. Usually normal glucose tolerance after delivery
4. 50% DM on long-term follow-up



Subjects at increased risk of DM

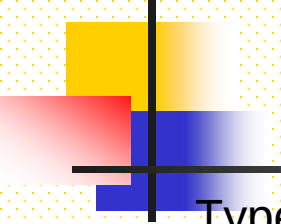
1. Prediabetes
2. DM in first degree relatives
3. History of gestational diabetes
4. Age over 35
5. Obesity, hypertension, dyslipidaemia



Comparison of clinical, genetic, and immunologic features of type 1 and type 2 diabetes

<u>Characteristic</u>	<u>Type 1</u>	<u>Type 2</u>
Prevalence	Caucasians: 10-20% Chinese: 5%	80-90% >95% of diabetic patients
Onset	Abrupt	Progressive/insidious
Endogenous insulin	Low to absent*	Normal/↑/↓
Ketosis	Common	Rare
Age at onset	Usu. children/young adults	Mostly in adults
Symptoms	Severe; dramatic wt. loss	May be none
Body mass	Usually non-obese	Obese or non-obese
Treatment	Insulin	Diet, oral drugs, insulin
Family history	10-15%	30%
Twin concordance	25-50%	70-90%
Human leukocyte antigen (HLA)	HLA-DR & DQ associations	Unrelated
Autoantibodies	Usu present at onset (>85%)	Absent

*check serum C-peptide



Pathogenesis of type 1 diabetes

Type 1 diabetes occurs as a result of islet cell destruction, immunologically mediated

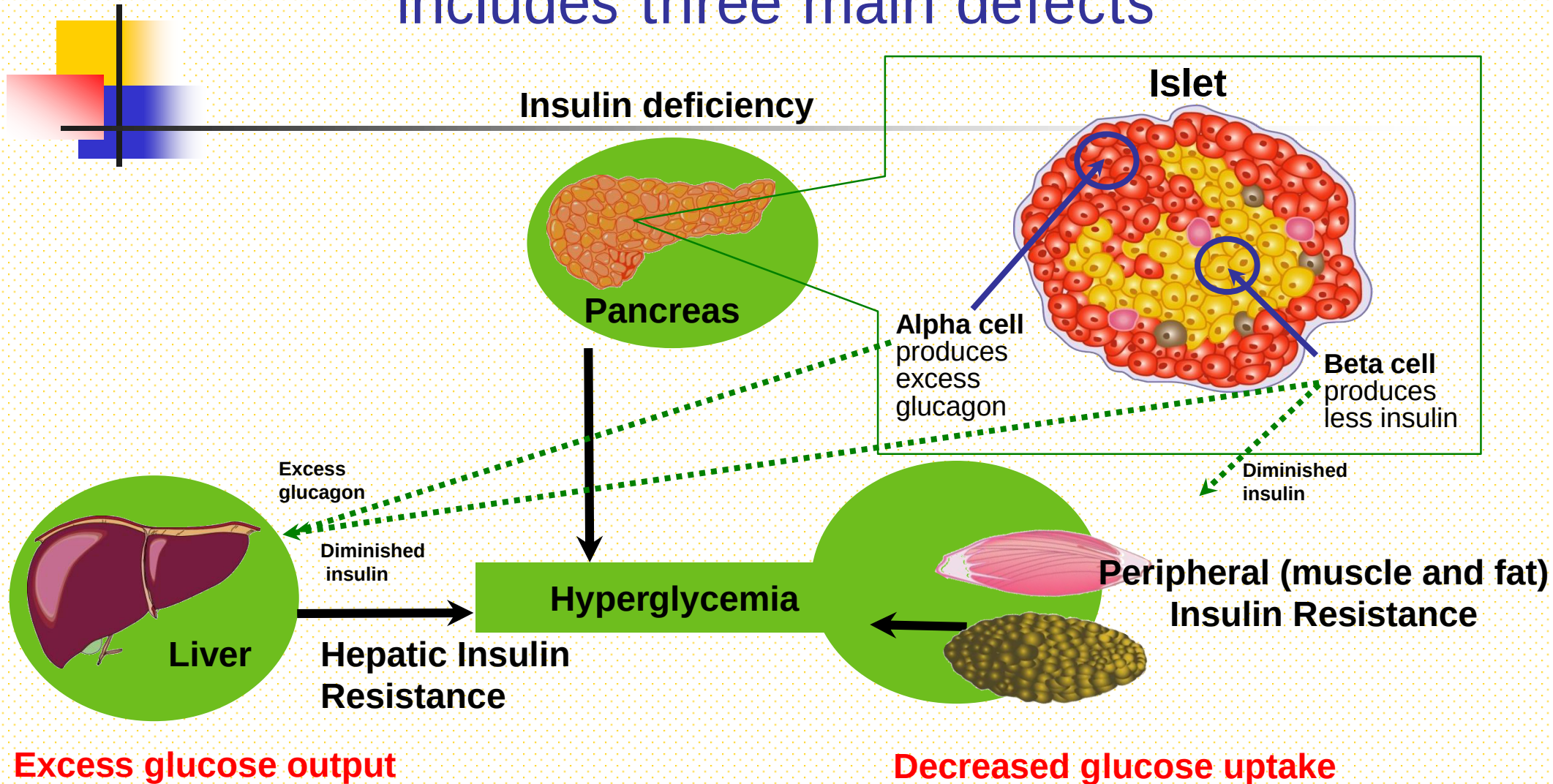
Genetic factors -Chromosome 6 DR/DQ susceptibility genes (20% risk in HLA-identical twins); other genes, eg. insulin gene

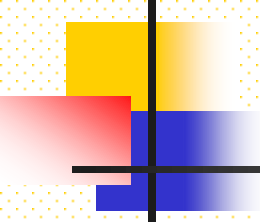
Environmental factors - exact nature uncertain, ? multi-hit action: virus (e.g. coxsackie B, mumps); Cow's milk protein in infancy

Immunological factors – evidenced by:

- a) Presence of islet autoantibodies (GAD; IA2; ZnT8; insulin) at onset
- b) Cell-mediated autoimmunity vs islet cells at autopsy of new onset patients
- c) Increased remission/ β cell preservation after immunosuppressants (eg. Cyclosporin/anti-TNF- α) in clinical trials on early cases
- d) Immunotherapy with Teplizumab, anti-CD3 antibody targetting T-cell, delayed onset of type 1 diabetes by 2 years in asymptomatic close relatives of type 1 diabetes, age >8 years, having at least two diabetes-related autoantibodies and at stage of prediabetes (*Herold KC et al, NEJM 2019*): FDA-approved Jan 2023

The pathophysiology of type 2 diabetes includes three main defects





Development & Progression of Type II Diabetes

Genes
Obesity, β -cell dysfunction,
Insulin resistance

+

Environment
Physical inactivity
Excessive food

Insulin resistance
Hyperinsulinaemia + **NGT**
(β -cell compensation)

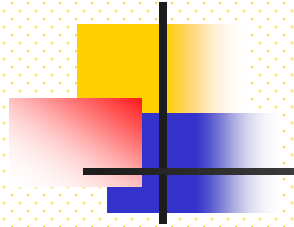


Beta cell decompensation

IGT



Progressive β -cell decline
Hypoinsulinaemia
Increasing hyperglycemia
Type 2 DM
(Decline accelerated by
Glucotoxicity & Lipotoxicity)



Complications : Acute

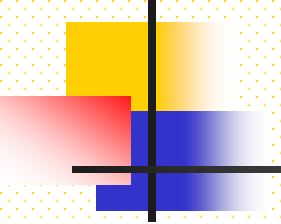
Diabetic ketoacidosis $\pm$ coma

Hyperosmolar nonketotic coma

Infections - pulmonary TB

urinary tract infection

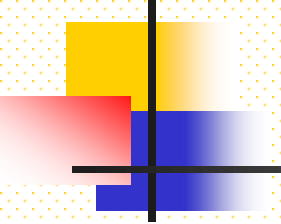
others



Hyperglycaemic Coma (Ketotic and Hyperosmolar non-ketotic)

Must exclude PRECIPITATING FACTORS:

- Infection; operation; trauma; severe emotional stress;
- Myocardial infarction; stroke (especially in the elderly)
- Drug administration e.g. steroid



Diabetic Ketoacidosis (Ketotic Coma)

1. First manifestation of undiagnosed Type 1 diabetes
2. Inadequate insulin treatment in known Type 1 diabetic patients

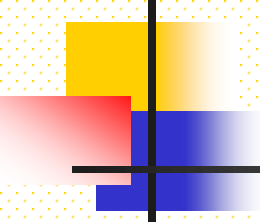
Acute insulin deficiency leads to unrestrained lipolysis, ↑ free fatty acids, excess hepatic ketone production and metabolic acidosis.

Hyperglycaemia causes osmotic diuresis, leading to dehydration and secondary loss of electrolytes

Insulin deficiency also causes impaired tubular sodium reabsorption

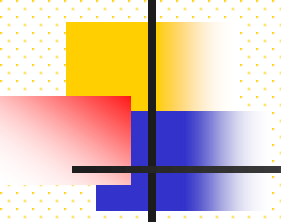
Net result: serum sodium and potassium usually normal on admission

Hypokalemia becomes evident after rehydration and insulin therapy (insulin stimulated glucose transport into cell is accompanied by inward K⁺ shift)



Treatment of diabetic ketoacidosis (DKA)

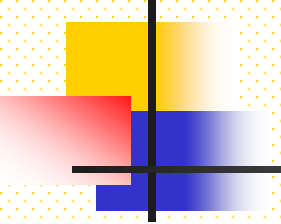
1. Replace salt and water deficit - IV normal saline + K^+ supplement (guided by serum K & renal function).
2. Insulin - low dose (5 - 10 u/h) by continuous IV infusion.
↓ rate when blood glucose < 13 mmol/L.
3. Correct acidosis - pH rises with rehydration and insulin.
Bicarbonate infusion if pH < 7.0.
4. Identify and treat the cause of DKA.
5. Monitoring of clinical response, BP, pulse, urine output; CVP and ECG as indicated.
6. Emergency measures/care of the unconscious patient.



Hyperosmolar non-ketotic diabetic coma

1. Usually undiagnosed type 2 diabetes
2. Insidious onset - days or weeks
3. Ketosis usually absent (just sufficient insulin to suppress lipolysis and ketogenesis; severe hyperosmolality may ↓ lipolysis)
4. Severe hyperglycaemia and dehydration
5. Severe hyperosmolality*
6. Depression of sensorium (when serum osmolality >340 mmol/L) with lack of thirst response
7. High mortality

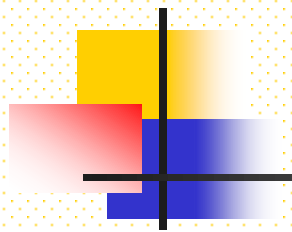
*The tendency to hyperosmolality may be worsened in elderly people, who may have reduced appreciation of thirst, or difficulty in drinking enough water to compensate for their osmotic diuresis, and may be on diuretics.



Treatment of hyperosmolar coma

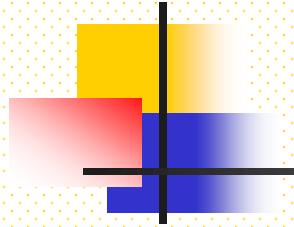
Same as for DKA except

1. In general, less insulin required.
2. Water deficit more severe.
3. Preferably, fluid replacement should be guided by CVP or pulmonary wedge pressure monitoring, especially if patient is elderly or has underlying cardiac or renal disease.
4. If Na^+ level > 150 mmol/L, use 1/2 normal saline.
5. Thrombosis is more common especially in Caucasians with hyperosmolality: clinical vigilance required.



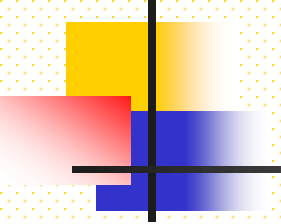
Hypoglycaemia

- Too much sulphonylurea or insulin;
- Too little carbohydrate;
- Too much exercise;
- Exclude renal impairment.



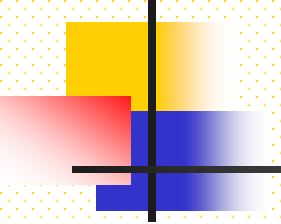
Symptoms of Hypoglycaemia

- Adrenergic:
palpitation; sweating; tremor
- Neuroglycopenic:
hunger sensation
restlessness & anxiety
numbness in fingers & around the mouth
altered consciousness; drowsiness & coma



Treatment of Hypoglycaemia

- Early meal or snack
- Sweet drinks, glucose tablets
- Intramuscular glucagon (1 mg)
- Intravenous glucose

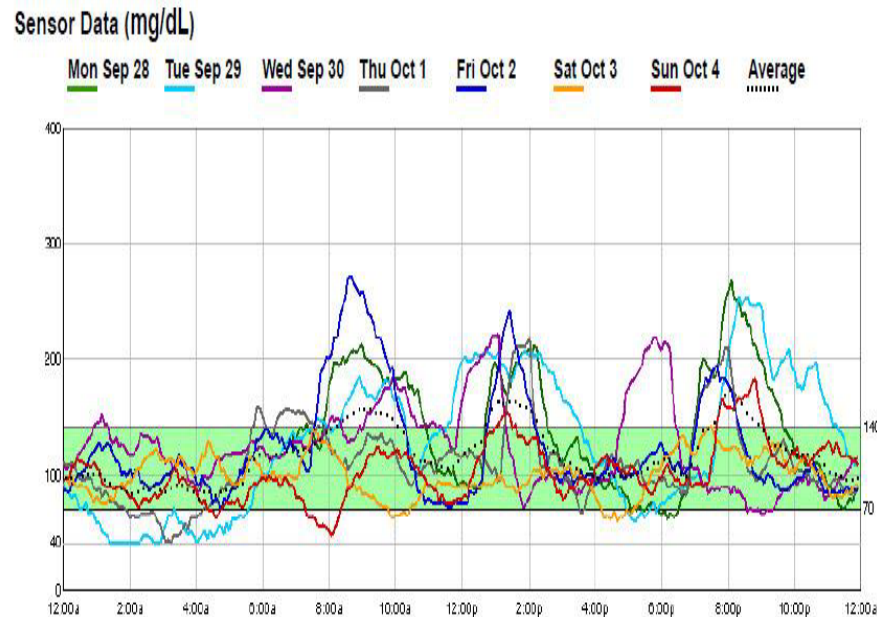


Monitoring Diabetic Control

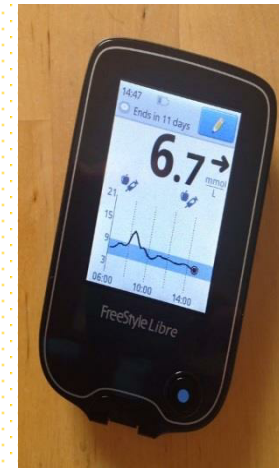
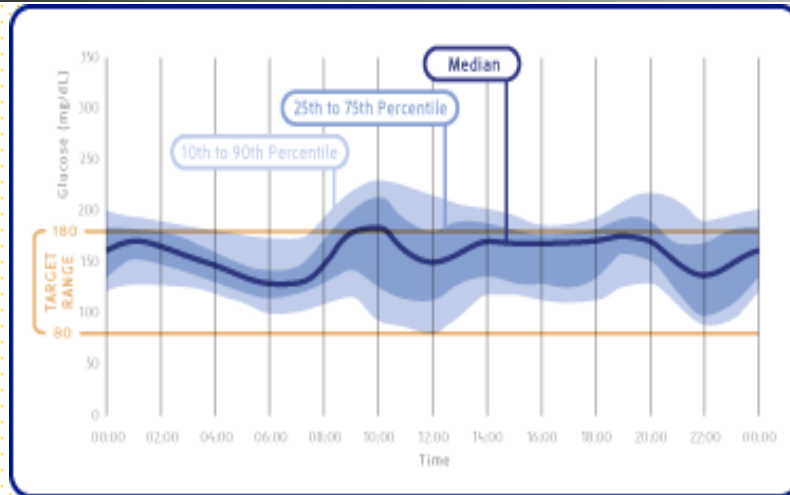
1. Home blood glucose monitoring including continuous glucose monitoring (CGMS)
2. Clinic blood glucose
3. Glycosylated haemoglobin / serum fructosamine

Continuous Glucose Monitoring (CGM): up to 15 days recording

Detection of hypoglycaemia unawareness; nocturnal hypoglycaemia and Somogyi phenomenon; post-prandial hyperglycaemia



Continuous Glucose Monitoring (CGM) Real Time CGM



Recent advances: direct reading/profile/summary (TIR etc)
via mobile APP; feed-back for pump adjustment; alarm system

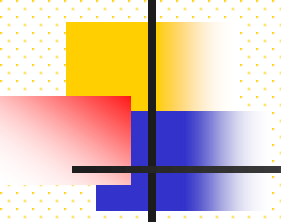
Treatment targets:

Time in range (TIR: 3.9-10mmol/L or 70-180mg/dl) => >70%

High (>10mmol/L) <25%; V high (>14mmol/L) <5%

Hypoglycaemia (<3.9mmol/L) <4%;

Serious hypoglycaemia (<3mmol/L) <1%



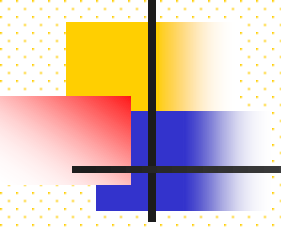
HbA1c Goals (ADA 2025)

Less than 7% in general

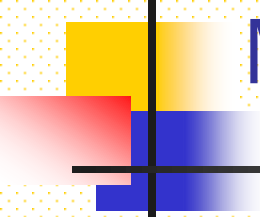
Individualization of treatment goals

Greater risk of hypoglycaemia at extremes of age

Less stringent target if multiple co-morbidities or short life expectancy

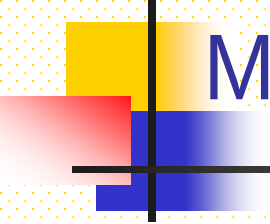


CHILDHOOD DIABETES



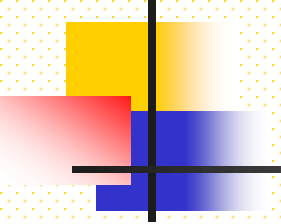
Metabolic Control of Diabetes in Childhood

- Type 2 diabetes in childhood: increasing prevalence due to obesity
 - Majority onset at 10-14 years; often asymptomatic (76% in HK)
 - More rapid deterioration of β -cell function and earlier development of diabetes-associated complications than adult-onset T2D
 - Target HbA1c $\leq 6.5\%$ (ISPAD 2024/ADA 2025)
 - Use GLP1-RA or SGLT2 inhibitors as additional therapy to metformin
 - Basal or basal-bolus insulin analogs if glycaemic target not reached; CGMS.
- Type 1 diabetes in childhood:
 - HK new-onset T1:T2 = 1.3:1 (Tung JY, Kwan EY, et al, Pediatr Diabetes 2022)
 - Basal-bolus insulin analogs; use of pump recommended; CGMS
 - Target HbA1c $\leq 7\%$; $\leq 6.5\%$ if on pump (ADA 2025)



Metabolic Control of Diabetes in Adolescence

- DCCT (Diabetes Control & Complications Trial): T1DM
- Benefits of good control (pump or multiple injections)
 - Primary prevention: retinopathy 53% ↓
microalbuminuria 10% ↓
 - Secondary prevention: retinopathy progression 70% ↓
microalbuminuria 70% ↓
 - Faster motor and sensory nerve conduction velocities
 - Lower total cholesterol levels
- Intensive therapy associated with 2-4 fold ↑ in severe hypoglycaemia (*J Pediatr* 1994)



Screening for Chronic Complications (Paediatrics Dept; QMH)

- Retinopathy, Nephropathy and Neuropathy-
including two first morning urine samples for microalbumin
- Annual thyroid dysfunction in T1 DM (3-5% patients may have autoimmune thyroid disease)
- Annual lipid levels
- T1 DM: Complication screening if (1) ≥ 5 years from diagnosis or (2) ≥ 2 years from diagnosis and ≥ 10 years old
- T2 DM: Start complication screening at diagnosis (HK: dyslipidaemia 35.3%, hypertension 22.5% and microalbuminuria 12.8% at diagnosis)

JC Whole Class Session

TELL US
WHAT
YOU THINK

THANK
YOU!



Look for today's teacher
and the others you may have
(HKU Portal login required)
missed.